GRADE 6 · CODING CLUB · TEACHER GUIDE

Course Plan

Build your own web API with FastAPI and Pydantic — 4 lessons for Grade 6

Kids go from "what is an API?" to running their own Pet Shelter API with automatic docs.

TODAY

1. Lesson overview
2. Learning goals
3. Setup checklist
4. Running the API
5. Assessment and differentiation

Lesson overview (4 × 60 minutes)

#	Lesson	Big idea	Kids build
1	**What is an API?**	Computers ask each other for things with **requests** and get **responses**, usually in **JSON**	A "human API" role-play, then read real JSON
2	**Pydantic: data with rules**	A **model** describes what good data looks like; Pydantic checks it and explains mistakes	A Monster Card model that rejects bad data
3	**My first FastAPI app**	A few lines of Python become a web API, with a free **/docs** page	A Fun Facts API with GET, path and query parameters
4	**Pet Shelter API**	FastAPI + Pydantic: **POST** new data and have it checked automatically	Add, list and find pets; showcase

TALKING POINTS

Each lesson = about 10 minutes of slides, 35–40 minutes hands-on, 10 minutes share and recap. Kids need basic Python first: variables, strings, numbers, functions, lists and dictionaries. If they haven't used dictionaries, spend 10 minutes on them before Lesson 1.

Learning goals

- Explain what an **API**, a **request** and a **response** are, using an everyday example
- Read and write simple **JSON**
- Describe data with a **Pydantic model**: types, defaults and simple rules (minimum, maximum, length)
- Read a **validation error** and fix the data
- Build a FastAPI app with **GET** and **POST** endpoints, **path** and **query** parameters
- Use the automatic **/docs** page to test an API
- Talk about why checking data matters (safety, fewer bugs)

TALKING POINTS

Language level: keep words simple and use the analogies on the slides (restaurant waiter, form with rules, robot receptionist). Vocabulary list for a word wall: API, request, response, JSON, endpoint, URL, GET, POST, model, type, validation, error, docs.

Setup checklist (do this before Lesson 1)

What	How	Check
Python 3.10 or newer	Install from python.org (tick "Add to PATH" on Windows)	`python --version`
FastAPI + Pydantic	`pip install "fastapi[standard]"` (Pydantic comes with it)	`fastapi --version`
A code editor	Thonny (simple, great for this age) or VS Code	Open and run a .py file
A web browser	Chrome, Edge, Firefox or Safari	Opens http://127.0.0.1:8000/docs
Starter folder	One folder per child, e.g. `my_api`, with an empty `main.py`	Kids can find it

TALKING POINTS

Everything runs on the school computer — no accounts or internet services needed, which is important because most online coding platforms require users to be 13 or older. If installs are blocked, ask IT to pre-install Python and run the pip command once per computer. Test the whole of Lesson 4 on one computer before the first lesson.

Running the API (what kids type)

Step	Command / action	What happens
1	Open a terminal in the `my_api` folder	Thonny: Tools → Open system shell
2	`fastapi dev main.py`	The API starts and reloads every time you save
3	Open `http://127.0.0.1:8000`	See the JSON response
4	Open `http://127.0.0.1:8000/docs`	The automatic interactive docs page
5	Press **Ctrl + C** in the terminal	Stops the API

TALKING POINTS

Print this table and stick it next to each computer. The most common problem is running the command in the wrong folder ("Could not find main.py"); the second is forgetting to save the file.

Assessment and differentiation

	Ideas
Exit tickets	Every lesson ends with 3 quick questions (answers in the talking points)
Lesson 4 showcase	Each child shows their API's /docs page and adds one pet live, then breaks it with bad data
Support	Give a printed starter file with blanks; pair kids up as "driver" and "navigator"; focus on GET only
Stretch	Extra rules with Field, a DELETE endpoint, a 404 error with HTTPException, more models (Owner, Adoption)
Success looks like	"I can explain what my API does, show it in /docs, and explain why it said no to bad data"